LINJI (JOEY) WANG

AI Robotics & Systems Engineer | Curriculum Learning & Embodied RL

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PROFILE

AI/robotics and systems engineer and Computer Science Ph.D. researcher specializing in automatic curriculum learning, deep reinforcement learning, and embodied-agent training. First/co-first author of two IROS 2025 papers, with systems experience spanning Aurora PostgreSQL query performance at AWS and C++/ROS 2 humanoid policy inference.

TECHNICAL EXPERTISE

Robot learning	Automatic curriculum learning, deep reinforcement learning, PPO, teacher–student learning, reward shaping, VAE task representations, sim-to-real
Robotics systems	PyTorch, Isaac Gym, Isaac Lab, MuJoCo, ROS 2, ONNX Runtime; navigation, quadruped locomotion, off-road mobility, humanoid policy inference
Systems & software	C, C++, Python, Aurora PostgreSQL, query processing and join optimization, AWS, Docker, Git/CI-CD

RESEARCH & ENGINEERING EXPERIENCE

RobotiXX Lab, George Mason University — **Fairfax, VA**
Graduate Research Assistant — Robot Learning — *Aug 2023 – Present*

- Developed **GACL**, an automatic curriculum-learning framework using VAE task representations, performance-history tracking, and target-domain grounding; trained PPO agents in 128 parallel Isaac Gym environments and improved success by 6.8% on wheeled navigation and 6.1% on quadruped locomotion versus state-of-the-art methods (first author, IROS 2025).
- Co-developed **Reward Training Wheels**, a teacher–student framework for proficiency-conditioned auxiliary-reward adaptation; in simulation, improved off-road mobility by 122.62% and reached the same performance threshold 3× faster than expert-designed rewards; sim-trained policies achieved 5/5 physical trials versus 2/5 (co-first author, IROS 2025).
- Third author on **Moving Through Clutter** (2026), a VR data-collection and evaluation framework for scene-aware humanoid locomotion with 348 trajectories across 145 3D scenes.
- Developing a C++/ROS 2 policy-inference stack for Unitree G1 with paired ONNX residual/base policies, metadata-driven observation construction, temporal history, normalization, and Isaac Lab–MuJoCo parity diagnostics.
- Co-authored RL-based **Adaptive Dynamics Planning** (4th author, ICRA 2026) and **Decremental Dynamics Planning** (3rd author, IROS 2025); the DDP-based RobotiXX system placed 2nd in both the simulation qualifier and physical finals of the 2025 BARN Challenge.

Amazon Web Services (AWS)
Software Development Engineer Intern — Amazon RDS & Aurora — *Summers 2025 & 2026*

- **Aurora PostgreSQL (2026):** Contributed to Adaptive Join, developing mechanisms that adjust join execution to improve query performance.
- Worked in the PostgreSQL-based database engine codebase in C on database internals, join processing, performance analysis, and compatibility improvements.
- **RDS Proxy (2025):** Built multi-region performance-comparison and regression-analysis tooling using Streamlit, CloudWatch metrics, statistical testing, and adaptive test selection.

Computational Engineering and Robotics Lab, Carnegie Mellon University — **Pittsburgh, PA**
Research Assistant — 3D Perception and AR — *Jan 2022 – May 2023*

- Built an AR scene-inpainting pipeline using GAN image completion plus RANSAC/DBSCAN point-cloud segmentation.

SELECTED PUBLICATIONS

- **Wang, L.**, Xu, Z., Stone, P., Xiao, X. "GACL: Grounded Adaptive Curriculum Learning with Active Task and Performance Monitoring." *IROS 2025.* First author.
- **Wang, L.***, Xu, T.*, Lu, Y., Xiao, X. "Reward Training Wheels: Adaptive Auxiliary Rewards for Robotics Reinforcement Learning." *IROS 2025.* Co-first author.
- **Additional robotics publications:** Moving Through Clutter (3rd author, arXiv 2026); DDP (3rd, IROS 2025); Adaptive Dynamics Planning (4th, ICRA 2026); ColorMap-VIO / II-NVM (5th, RA-L 2026 / 2025).

EDUCATION

George Mason University — Fairfax, VA
Ph.D. in Computer Science — research focus: curriculum learning and reinforcement learning for robotics — *2023 – Present*

Carnegie Mellon University — Pittsburgh, PA
M.S. in Mechanical Engineering, GPA: 3.94/4.0 — *2021 – 2023*

University of Cincinnati — Cincinnati, OH
B.S. in Mechanical Engineering, magna cum laude — *2016 – 2021*